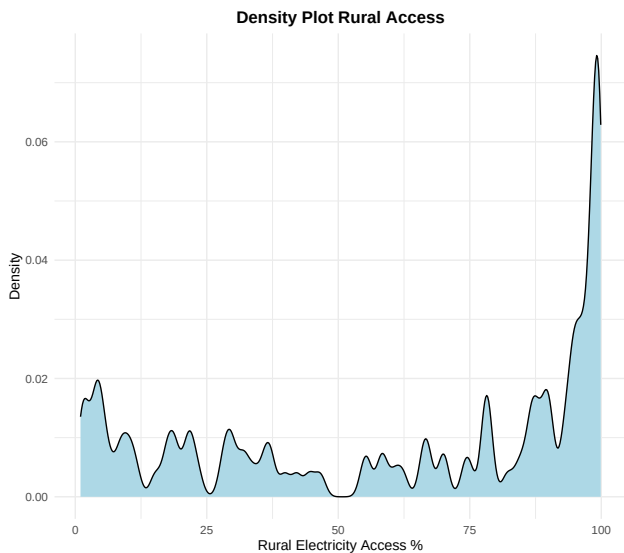


Government Effectiveness and Rural Electrification: A Cross-Sectional Analysis on Determinants of Successful Grid Expansion

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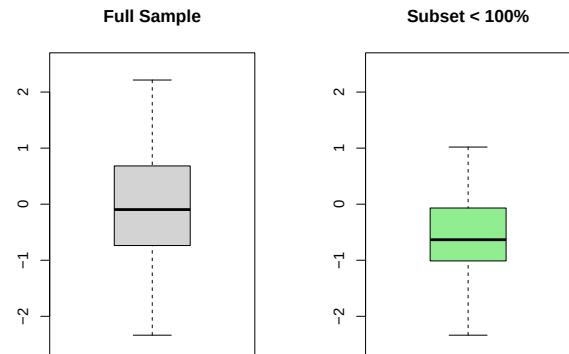
Abstract

This paper investigates whether the quality of governance has a significant effect on the electrification of rural households in countries where rural electrification is not yet complete. The central motivation draws on the observation that extending electricity access to remote, sparsely populated areas is rarely commercially profitable for private actors, as low population density and limited purchasing power suppress expected returns on grid infrastructure investment. In the absence of market incentives, targeted state intervention and effective government programs therefore become the primary mechanism through which rural populations gain electricity access. The analysis connects this practical observation to a broader hypothesis: that targeted, effective policies are a key structural determinant of rural electricity access in countries that have not yet achieved universal coverage, and that cross-country variation in electrification rates can in meaningful part be traced back to differences in governance quality.



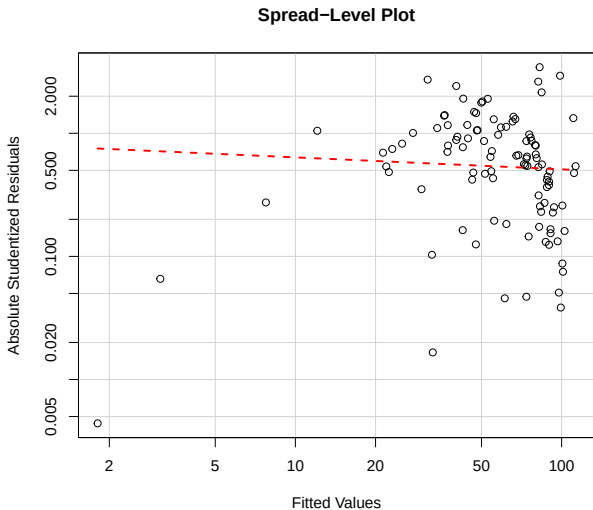
The empirical analysis uses cross-sectional data from the World Bank World Development Indicators (WDI) for the year 2016, the most complete available year for the dependent variable, covering up to 217 countries across 11 variables. The dependent variable is the share of rural households with some form of electricity access. The sample is restricted to countries with rural electrification below 100%, yielding 123 relevant observations, as countries with complete coverage face a structural ceiling that prevents any further increase regardless of governance quality, wealth, or other relevant factors, which would introduce a mechanical truncation bias into estimated coefficients. The primary independent variable is the World Bank Government Effectiveness Score, a composite indicator ranging from -2.5 to $+2.5$ constructed from 30 sub-indicators capturing the quality of public services, the degree of civil service independence from political pressures, the quality of policy formulation and implementation, and the credibility of government commitment to stated poli-

cies. This is particularly relevant for energy policy and electricity grid expansion because large-scale energy infrastructure projects depend heavily on effective state capacity to coordinate long-term planning, enforce regulatory frameworks, attract private and public investment, manage complex permitting and land acquisition processes, and ensure the reliable implementation of national electrification and transmission strategies.



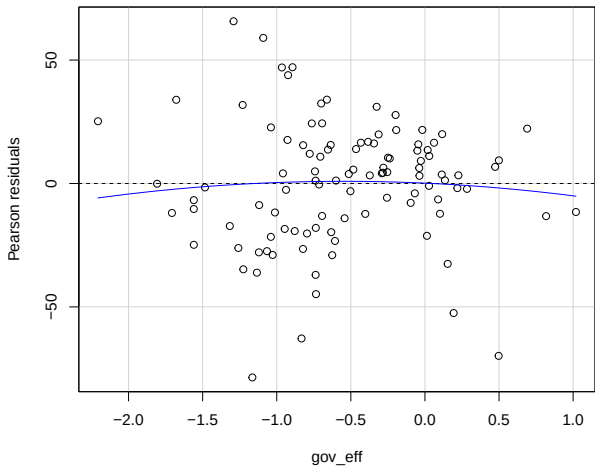
A positive bivariate correlation of approximately 0.56 between government effectiveness and rural electrification provides preliminary support for the hypothesis prior to any regression analysis. The initial bivariate regression model establishes a positive and statistically significant baseline relationship between government effectiveness and rural electrification ($\hat{\beta} = 31.27$, $p < 0.01$, $R^2 = 0.317$). The coefficient implies that a one-unit increase in the governance score is associated with approximately 31.27 percentage points of additional rural electrification, or roughly 3.13 percentage points per 0.1-unit increase. Government effectiveness alone accounts for approximately 31% of the variation in rural electrification across the restricted sample, a substantively meaningful result for a single-variable model. Two control variables are subsequently introduced in separate multivariate specifications to assess the robustness of this baseline finding. First, electricity transmission losses as a share of total output are included as a proxy for overall grid infrastructure quality and maintenance. This is based on the reasoning that governments unable to sustain an efficient existing grid may be equally unable to extend that grid to rural peripheries. High transmission losses may also reflect broader institutional deficiencies, including weak regulatory oversight, underinvestment in maintenance, limited technical capacity, and ineffective utility governance, all of which can constrain a state's ability to successfully implement grid extension programs. This variable enters with a negative but statistically insignificant coefficient and marginally reduces the adjusted R^2 relative to the bivariate model, suggesting that transmission losses do not independently confound the relationship between governance and rural access. The reduction in sample size by 30 observations when in-

cluding this variable further limits its analytical value. Second, log-transformed GDP per capita adjusted for purchasing power parity is introduced, motivated by the widely supported theoretical expectation that wealthier countries have greater fiscal capacity to finance costly grid extensions and rural electrification programs. The variable captures expected profitability of grid extensions, as lower adjusted GDP per capita signals lower demand for electricity due to limited financial means of citizens. The inclusion of logGDP substantially attenuates the government effectiveness coefficient to 9.99 ($p < 0.1$) while GDP itself is highly and robustly significant ($\hat{\beta} = 24.79$, $p < 0.01$), with the adjusted R^2 rising markedly to 0.514. This pattern is consistent with the theoretical argument advanced by Acemoglu and Robinson (2012), who contend that institutional quality and economic prosperity are deeply intertwined, suggesting that part of the apparent governance effect in the bivariate model operates indirectly through the income channel. An interaction term between government effectiveness and rural population share, introduced to capture the possibility that governance matters more where the rural electrification challenge is quantitatively larger, does not yield significant results and leaves adjusted R^2 essentially unchanged, indicating that the modifying role of rural population size is not detectable in this cross-section. Regression assumption tests for the preferred multivariate model reveal a mildly curved residual pattern against both regressors, indicating a possible but not severe violation of the Zero Conditional Mean assumption. The Breusch-Pagan test returns a p -value of 0.062, suggesting marginal heteroskedasticity at the 10% significance threshold.



Implementing HC1-robust standard errors in response causes government effectiveness to lose significance entirely, while logGDP remains significant at the 1% level throughout all specifications. Given that the Breusch-Pagan result is itself only

marginally significant, the non-robust model is retained as the preferred specification, accepting a minor efficiency trade-off to preserve interpretability.



The results collectively indicate that GDP per capita is the dominant explanatory variable for rural electrification among countries that have not yet reached universal coverage, while government effectiveness exerts a positive but fragile independent effect that diminishes substantially once economic wealth is controlled for. No OLS assumptions are directly violated in a manner that would invalidate the model’s core findings. The analysis further identifies a potential missing data pattern consistent with selection bias: countries with government effectiveness scores below the sample mean account for 10% of missing observations in the rural electrification variable, compared to only 2% among better-governed countries, suggesting that the current sample likely over-represents well-functioning states and may understate the true effect of poor governance on rural energy deprivation. Future work should draw on panel data with greater temporal coverage to exploit within-country variation, incorporate more targeted proxies for energy-specific government capacity such as public energy expenditure or average outage frequency, and address missing data patterns that reduce the representativeness of the current cross-section. The findings nonetheless provide meaningful cross-sectional evidence that governance quality is a structurally relevant factor in the global rural electrification process.

Key References

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